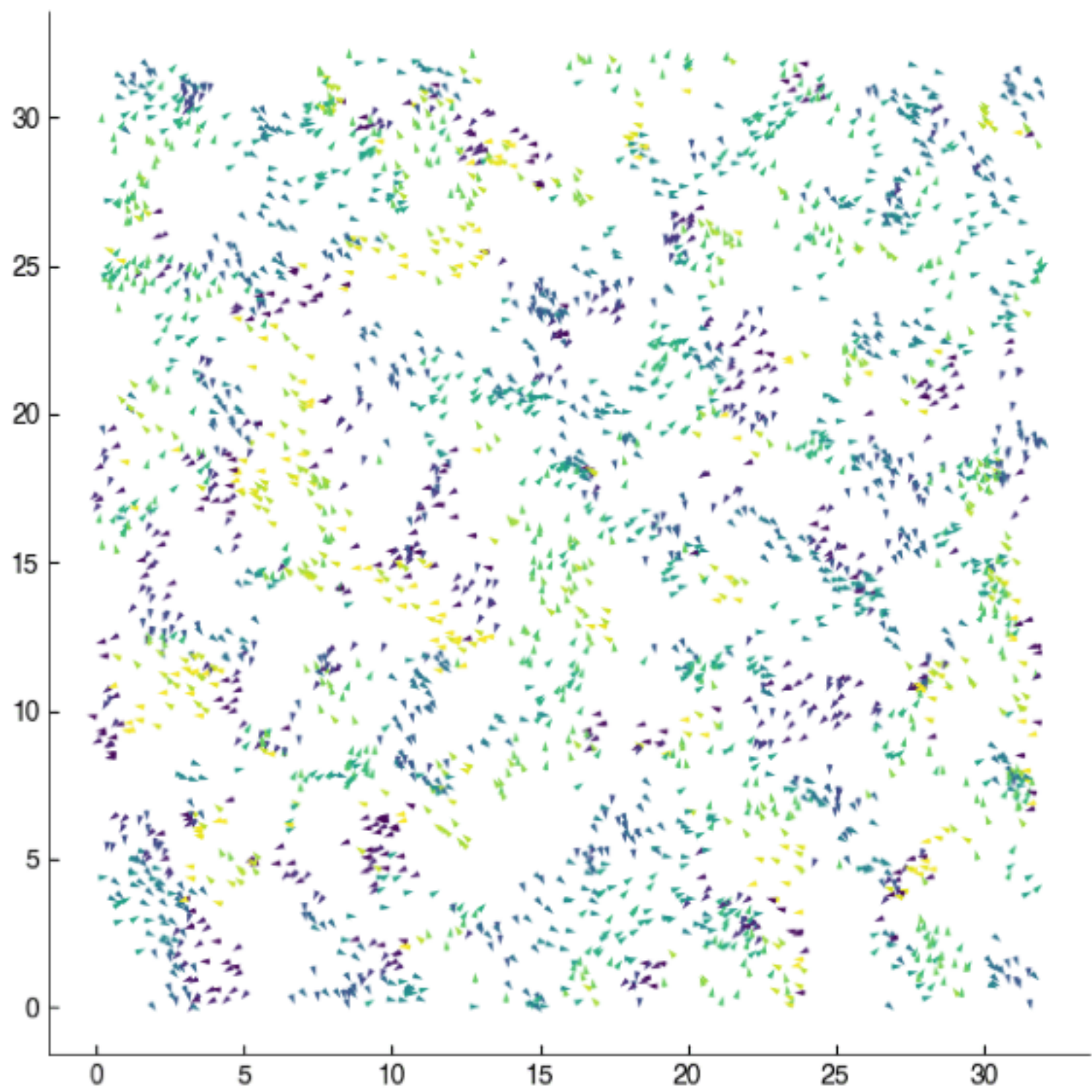


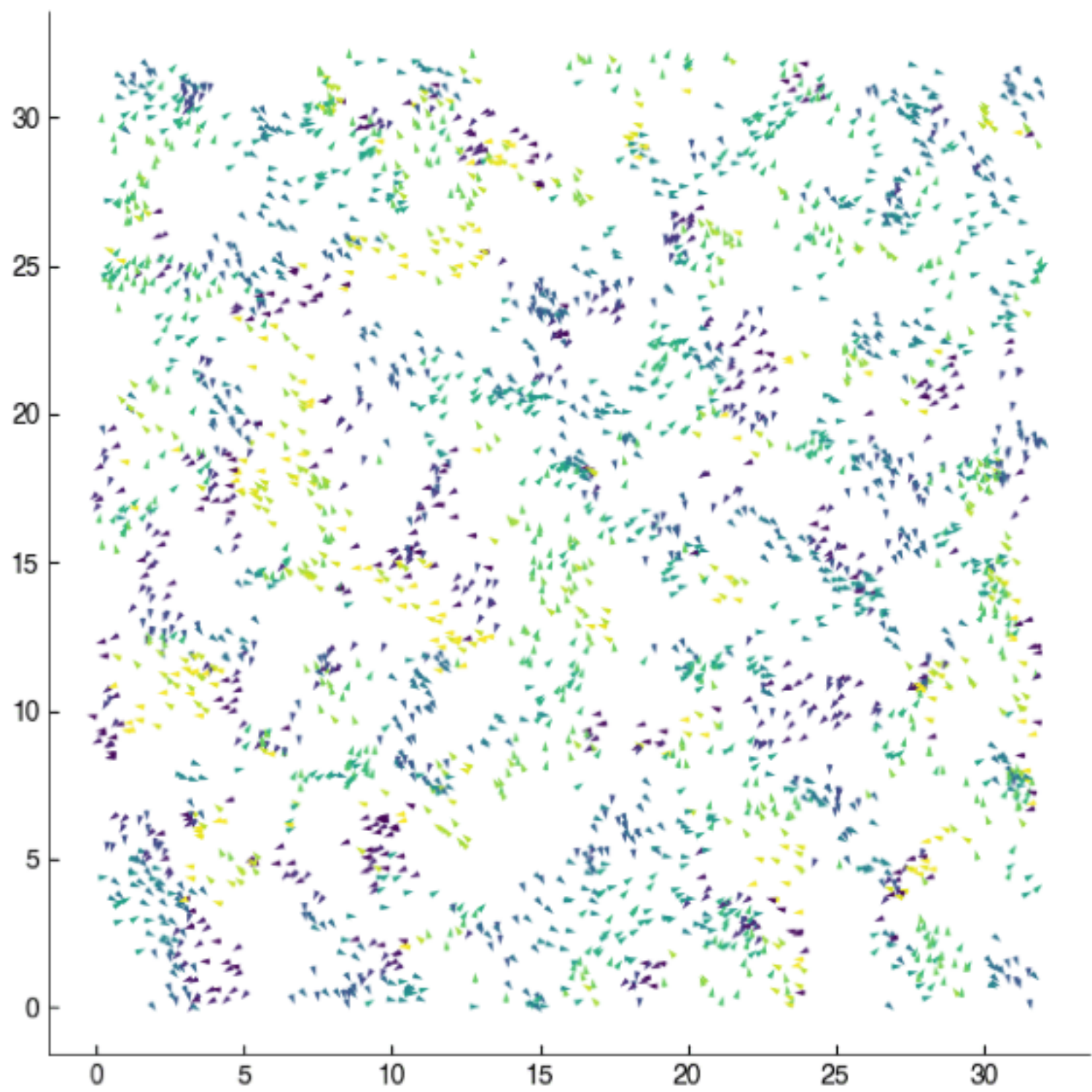


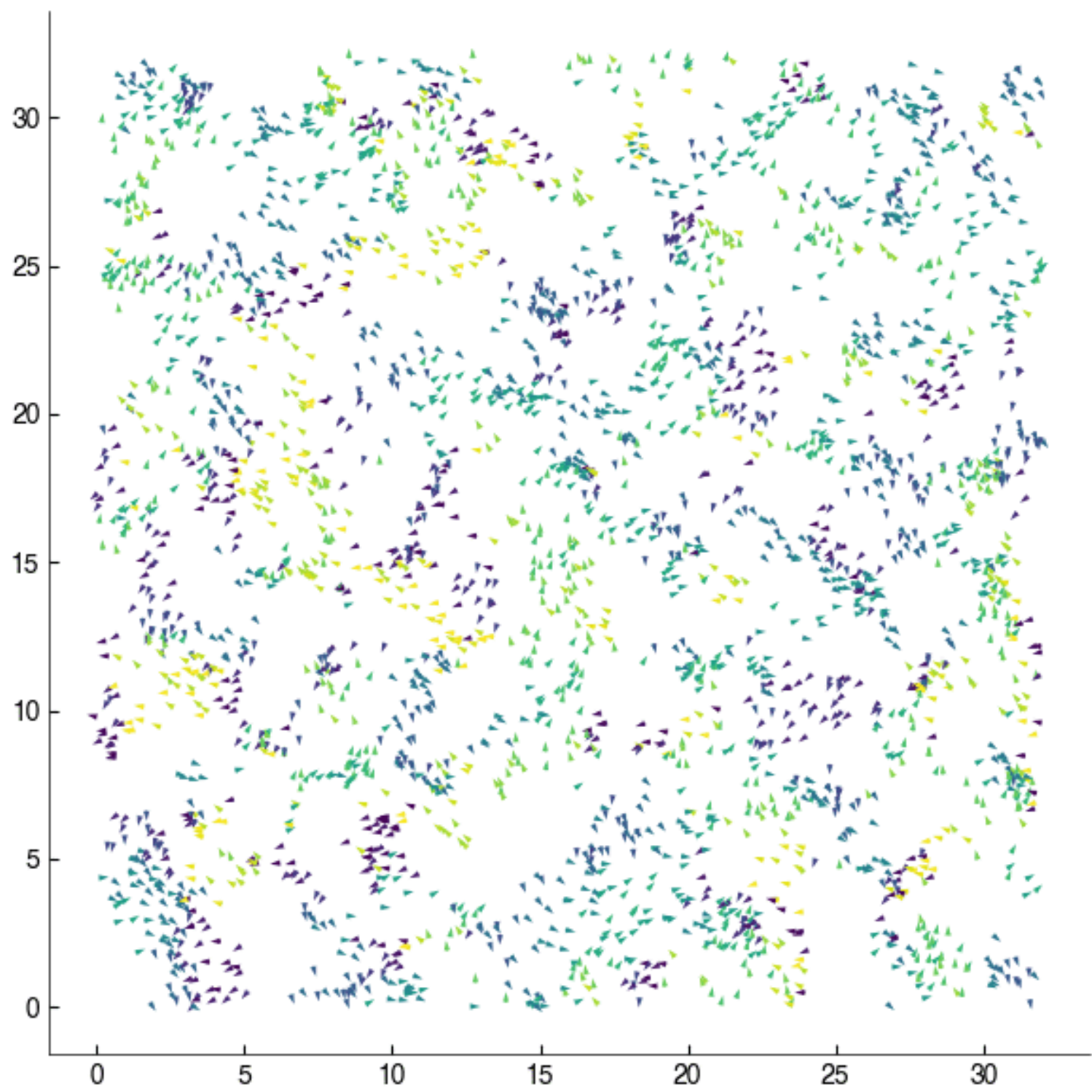
Vicinity Simulation F. Turki 2020

$$\begin{cases} dX_t^i &= P \operatorname{ev}(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \chi F\left((X_t^i, \theta_t^i), \mu_t^N\right) dt + \sqrt{2} dB_t^i \end{cases}$$

- Interaction strength $\chi > 0$
- Empirical measure: $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{(X_t^i, \theta_t^i)}$, McKean-Vlasov
- Famous interacting active Brownian particle model: Vicsek model (Phys. Rev. Lett., 1995) for flocking birds, 6000+ citations

Interacting ants

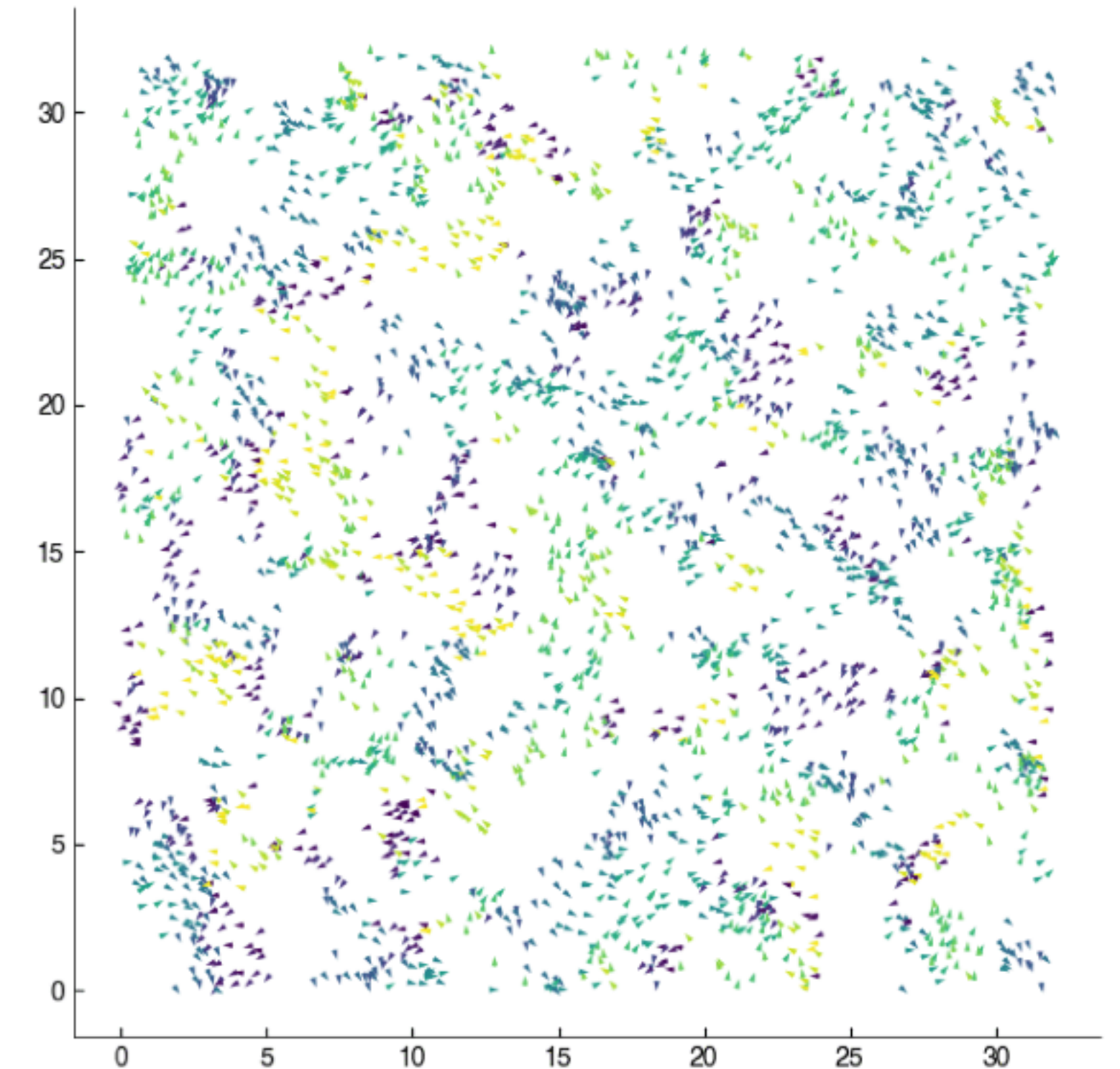




Interacting ants

$$\begin{cases} dX_t^i &= \text{Pev}(\theta_t^i)dt + \sqrt{2\sigma_x}dW_t^i \\ d\theta_t^i &= \chi F\left((X_t^i, \theta_t^i), \mu_t^N\right)dt + \sqrt{2}dB_t^i \end{cases}$$

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Vicsek model simulation © F. Turci 2020

Interacting ants

$$\begin{cases} dX_t^i &= \text{Pev}(\theta_t^i)dt + \sqrt{2\sigma_x}dW_t^i \\ d\theta_t^i &= \chi F\left((X_t^i, \theta_t^i), \mu_t^N\right) dt + \sqrt{2}dB_t^i \end{cases}$$

- Ant model (dW, Bruna & Burger, SIAM J. Appl. Dyn. Syst., 2025):
$$F = n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j)$$